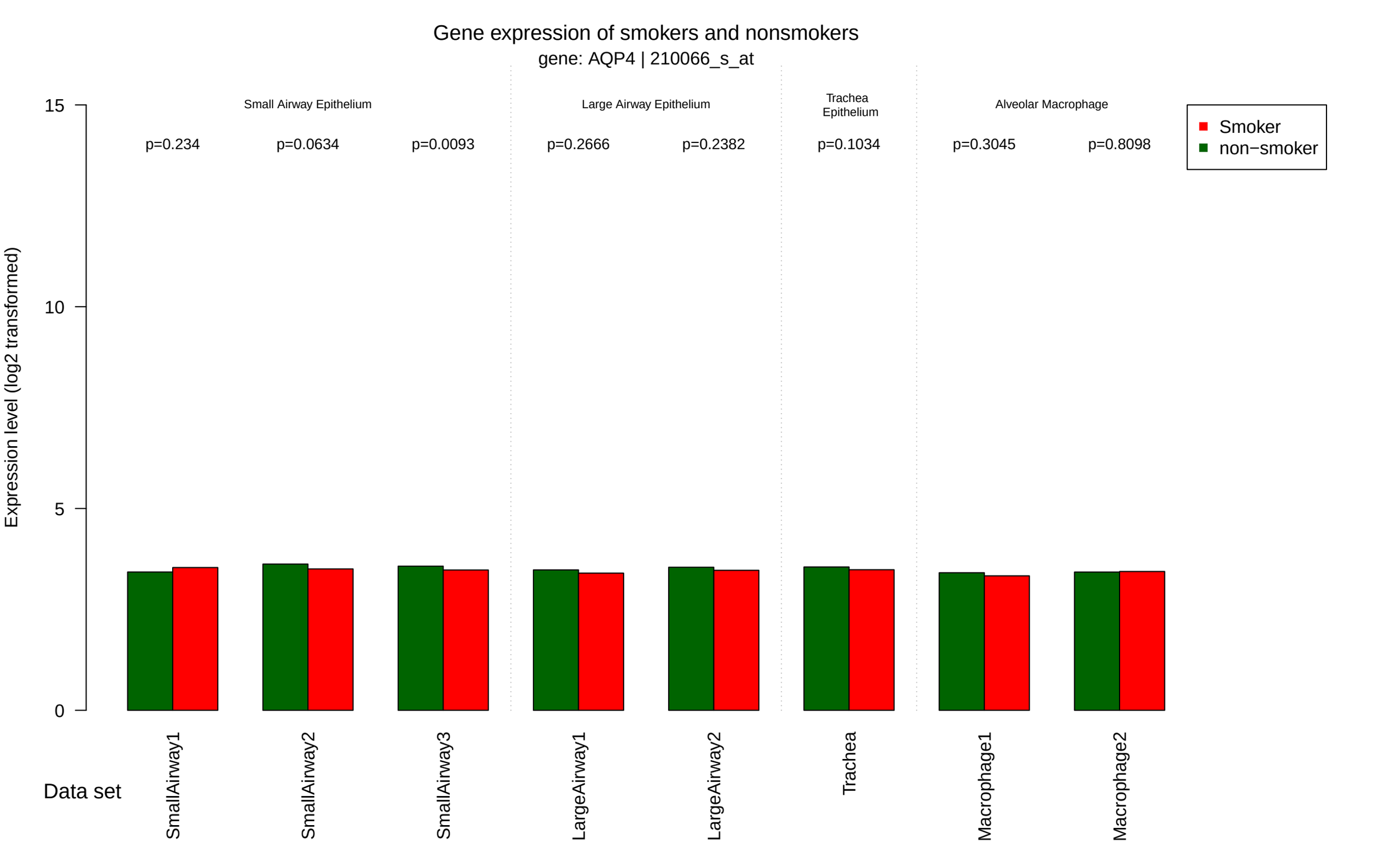
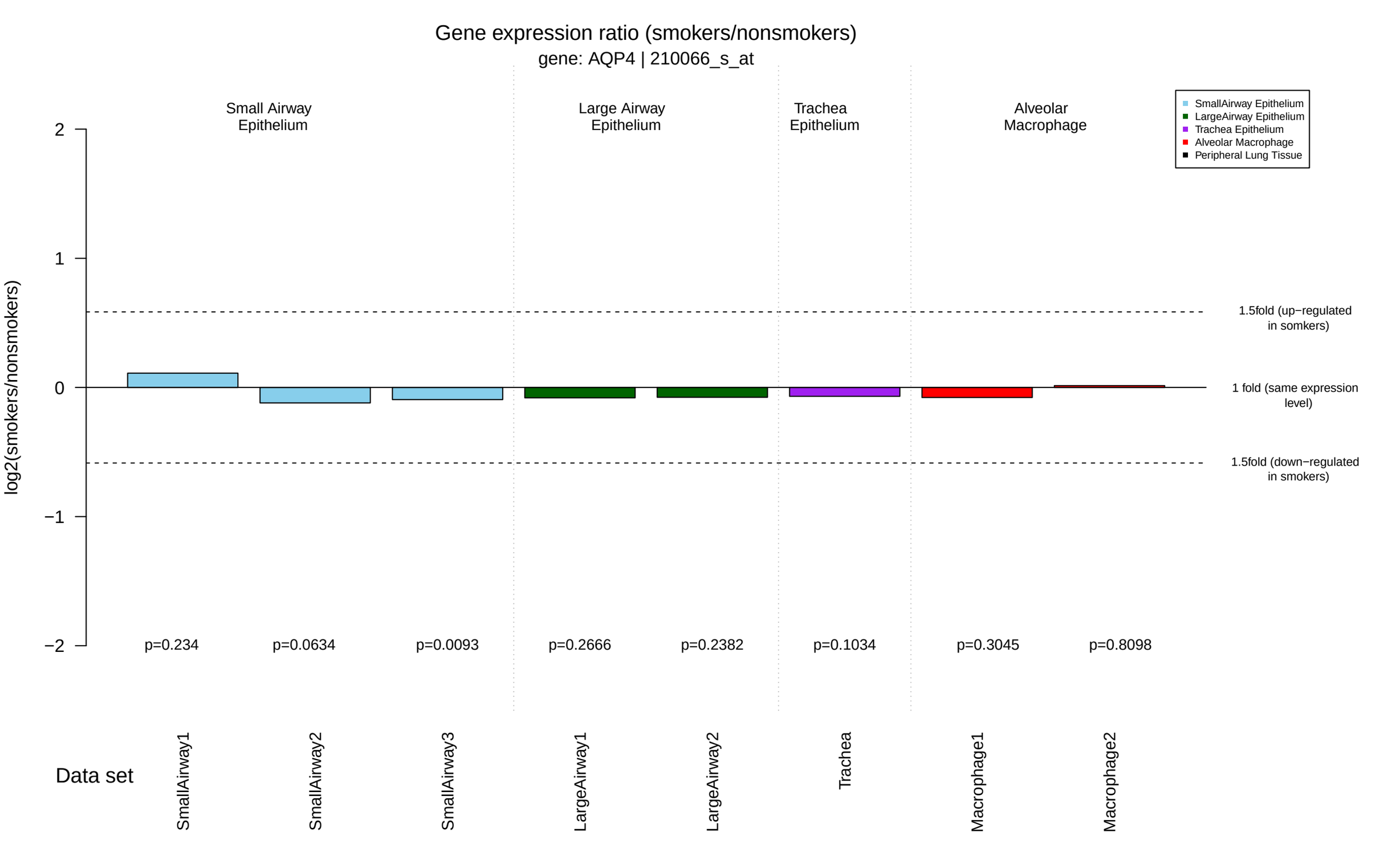
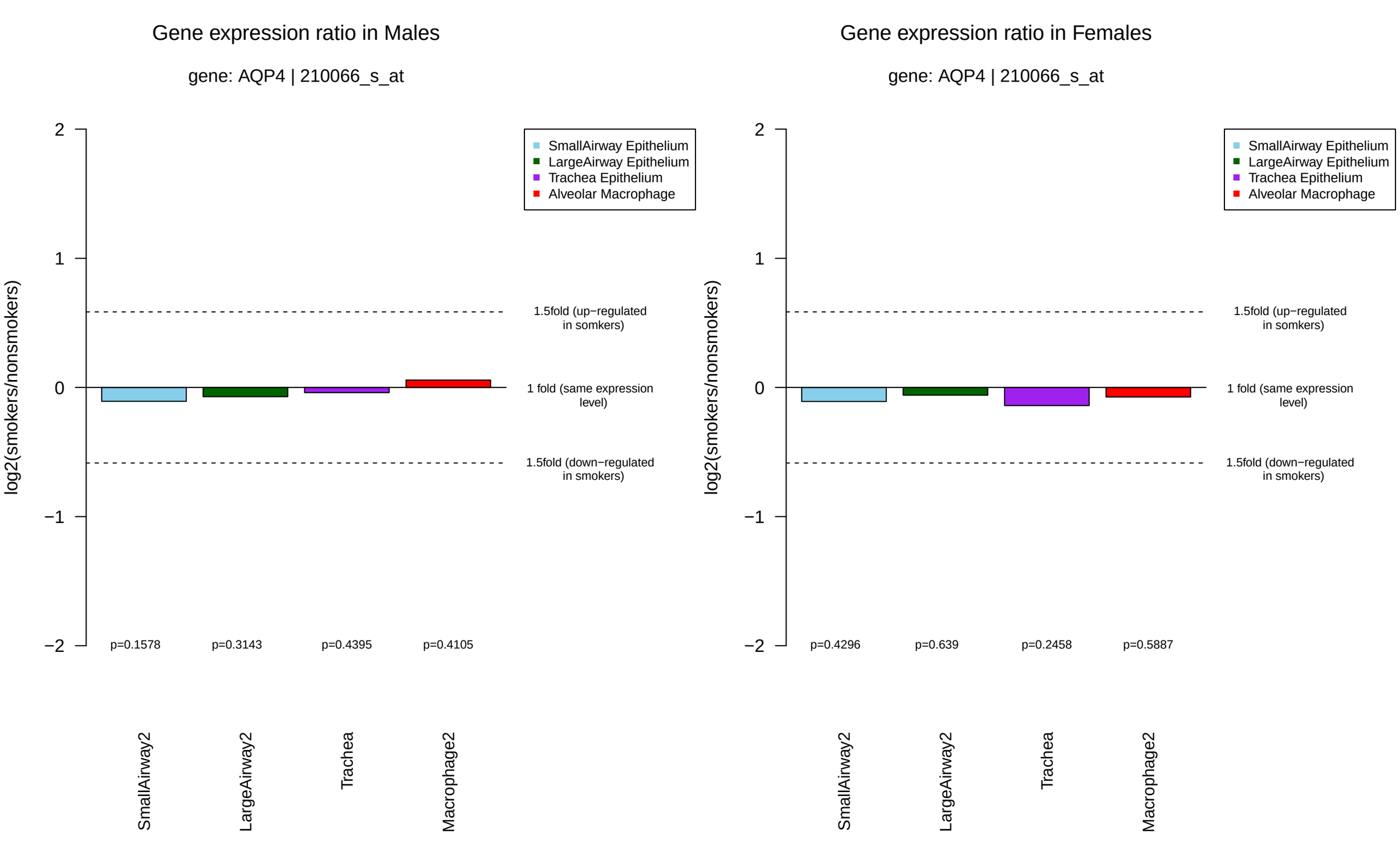
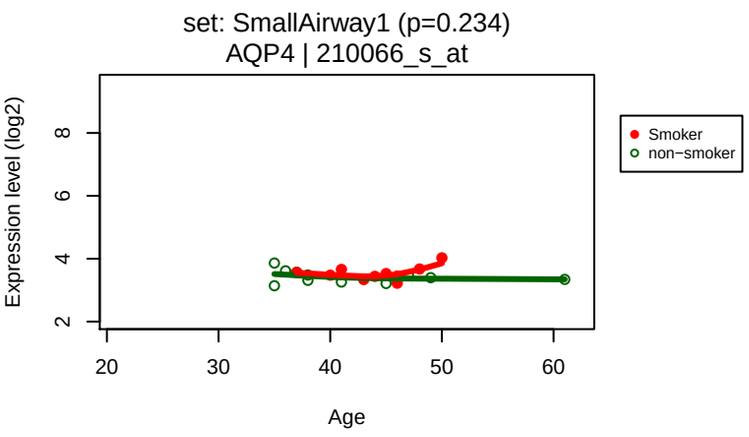
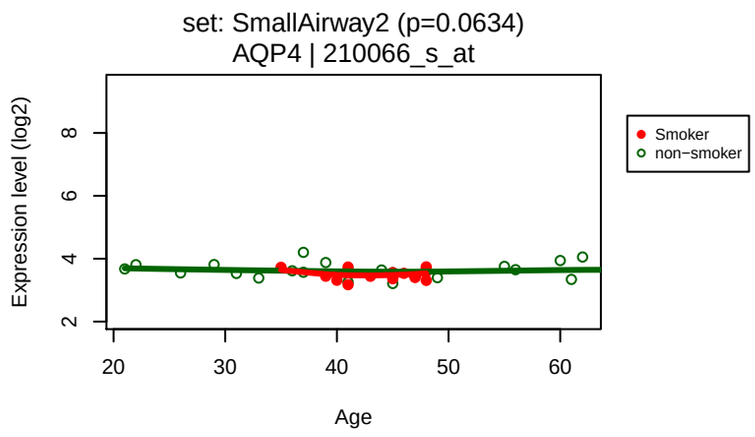
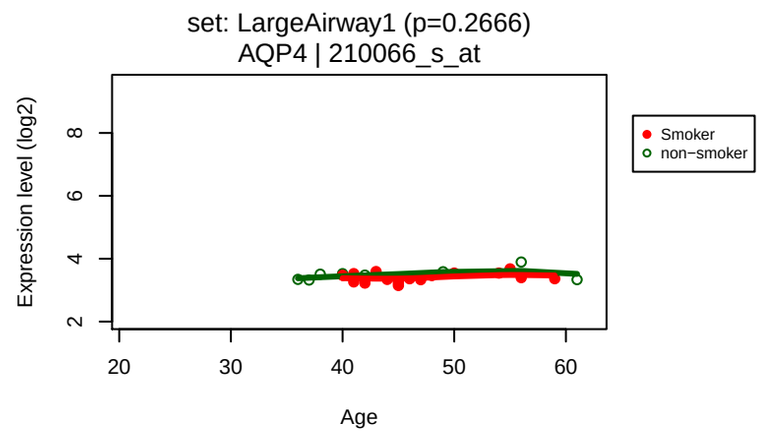

Gene expression of smokers and nonsmokers
gene: AQP4 | 210066_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.234
p=0.0634
p=0.0093
p=0.2666
p=0.2382
p=0.1034
p=0.3045
p=0.8098
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

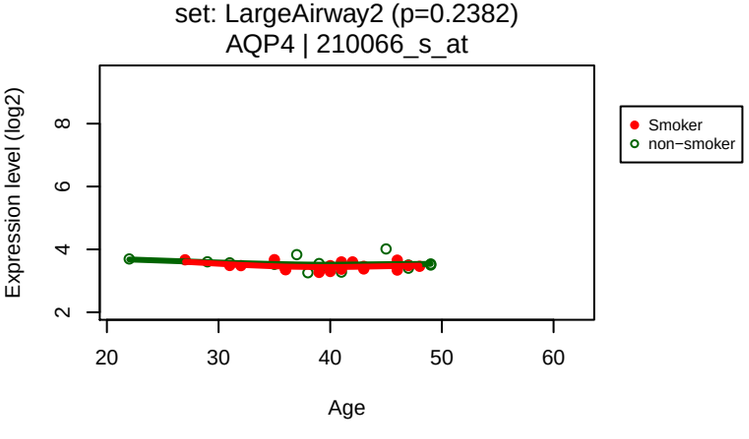

Gene expression ratio (smokers/nonsmokers)
gene: AQP4 | 210066_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.234
p=0.0634
p=0.0093
p=0.2666
p=0.2382
p=0.1034
p=0.3045
p=0.8098
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

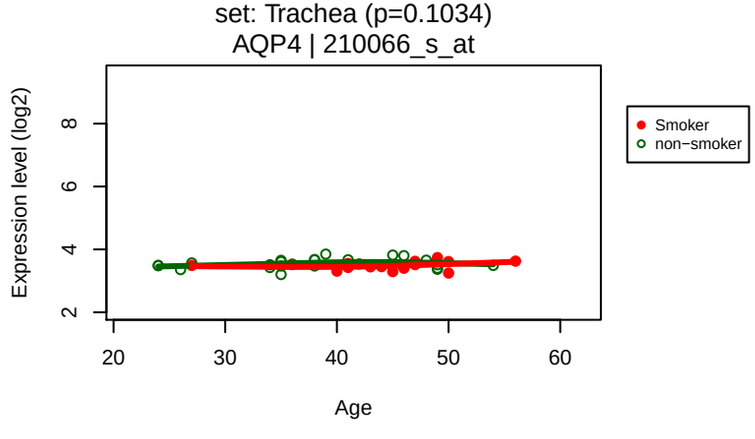

Gene expression ratio in Males
gene: AQP4 | 210066_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1578
p=0.3143
p=0.4395
p=0.4105
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AQP4 | 210066_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4296
p=0.639
p=0.2458
p=0.5887
SmallAirway2
LargeAirway2
Trachea
Macrophage2

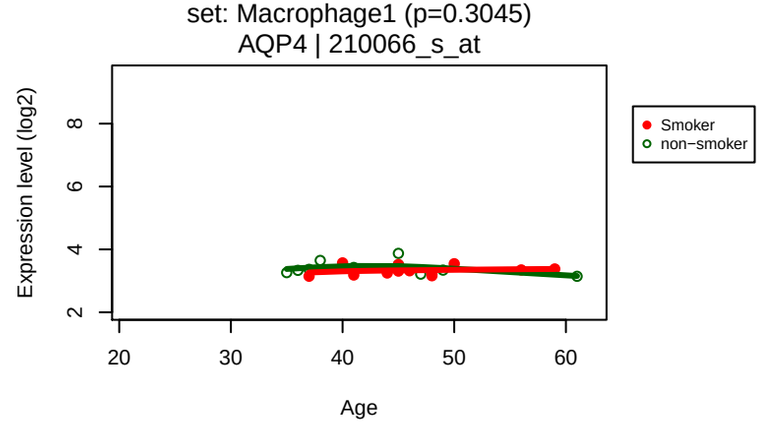

set: SmallAirway1 (p=0.234)
AQP4 | 210066_s_at
Expression level (log2)
Age
Smoker
non−smoker

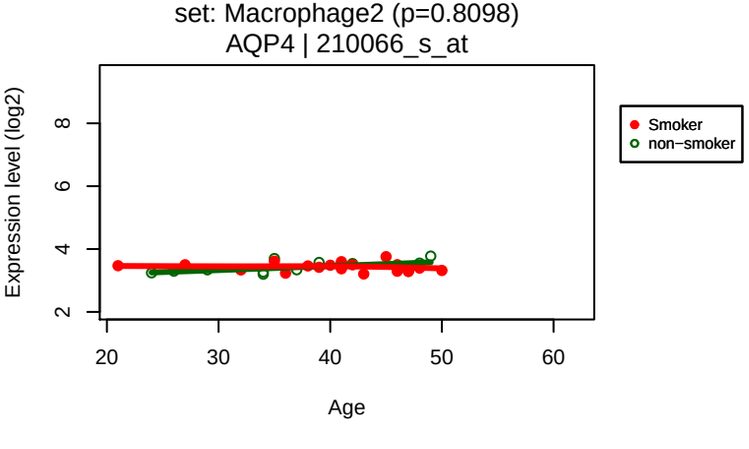

set: SmallAirway2 (p=0.0634)
AQP4 | 210066_s_at
Expression level (log2)
Age
Smoker
non−smoker

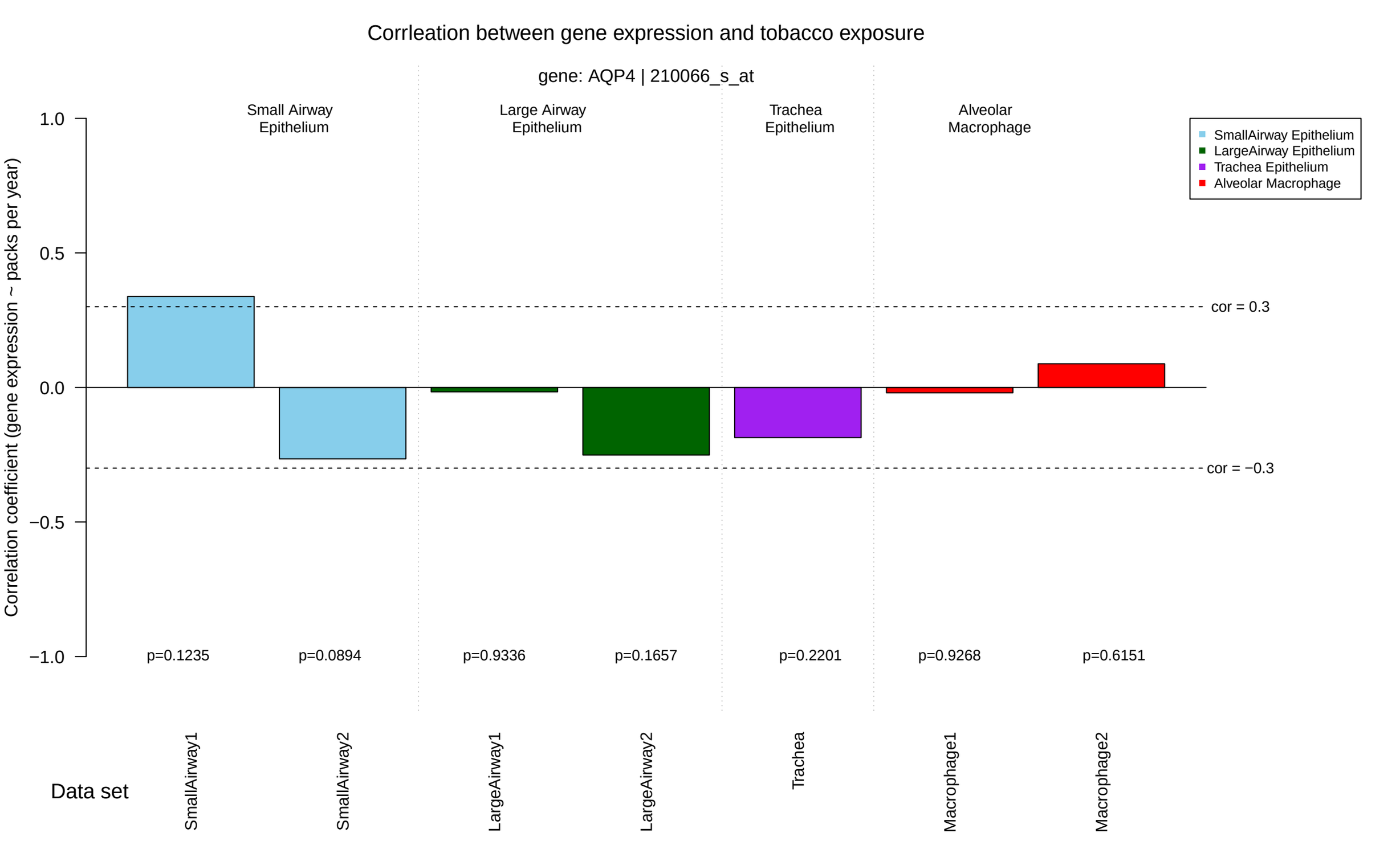

set: LargeAirway1 (p=0.2666)
AQP4 | 210066_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2382)
AQP4 | 210066_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1034)
AQP4 | 210066_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3045)
AQP4 | 210066_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8098)
AQP4 | 210066_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AQP4 | 210066_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1235
p=0.0894
p=0.9336
p=0.1657
p=0.2201
p=0.9268
p=0.6151
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2